5. Results: Neuron-Level Editing (FiNE)

Implementation on Pythia-6.9B

We implemented a full FiNE-style editing framework matching the official formulation:

- **Localization:** Uses FiNE score c_i^ℓ to identify Top-K neurons in target layers (e.g., Layer 28).
- **Patching:** Attaches learnable patch vectors via forward hooks to FFN output weights.
- **Optimization:** Minimizes a multi-term loss balancing edit success (L_{edit}), preservation of old behavior (L_{old}), KL divergence, and penalties.

Experimental Findings

We ran FiNE on the 50 high-overlap (SET-POLY) and 50 low-overlap (SET-MONO) pairs. We measure **Locality** (accuracy on Fact B) and **Avg. Drop** (clipped mean percentage drop in Fact B probability).

Group (N=50)	Edit Success (A)	Locality (B Acc.)	Avg. Drop (P_B)
SET-POLY (High)	12%	**10%** (was 88%)	**88.47%**
SET-MONO (Low)	58%	**24%** (was 82%)	**77.20%**

Table 1: FiNE neuron-level editing results. High overlap leads to near-complete erasure of Fact B. Surprisingly, even low-overlap controls suffer severe probability drops.

Analysis of FiNE Performance

- **Severe Bleed-over in High Overlap:** In SET-POLY, FiNE frequently devastates Fact B, with accuracy dropping to 10%. The high overlap creates a "bottleneck" where updates sharp enough to affect A inevitably slice through B.
- **FiNE is Intrinsically "Sharp":** The crucial finding is that even in the low-overlap SET-MONO group, FiNE causes a massive **77.20% average drop** in Fact B probability. This indicates that with current hyperparameters, FiNE's updates are aggressively non-local, altering the representation space significantly even when targeted neurons are mostly disjoint.